

MODEL-VIEW-CONTROLLER (MVC) PATTERN**Student.java:**

```
public class Student {
    private String name;
    private int id;
    private String grade;

    public Student(String name,int id,String grade){
        this.name=name;
        this.id=id;
        this.grade=grade;
    }

    public String getName(){
        return name;
    }

    public void setName(String name){
        this.name=name;
    }

    public int getId(){
        return id;
    }

    public void setId(int id){
        this.id=id;
    }

    public String getGrade(){
        return grade;
    }

    public void setGrade(String grade){
        this.grade=grade;
    }
}
```

StudentController.java:

```
public class StudentController {
    private Student student;
    private StudentView view;

    public StudentController(Student student, StudentView view){
        this.student=student;
        this.view=view;
    }

    public void setStudentName(String name){
        student.setName(name);
    }

    public void setStudentId(int id){
        student.setId(id);
    }

    public void setStudentGrade(String grade){
        student.setGrade(grade);
    }

    public String getStudentName(){
        return student.getName();
    }

    public int getStudentId(){
        return student.getId();
    }

    public String getStudentGrade(){
        return student.getGrade();
    }

    public void updateView(){
        view.displayStudentDetails(student.getName(),student.getId(),student.getGrade());
    }
}
```

StudentView.java:

```
public class StudentView {  
    public void displayStudentDetails(String name,int id,String grade){  
        System.out.println("Student Details");  
        System.out.println("Name: "+name);  
        System.out.println("ID: "+id);  
        System.out.println("Grade: "+grade);  
    }  
}
```

MVCPatternTest.java:

```
public class MVCPatternTest {  
    public static void main(String[] args){  
        Student student=new Student("Perna",101,"A");  
        StudentView view=new StudentView();  
  
        StudentController controller=new StudentController(student,view);  
  
        controller.updateView();  
  
        System.out.println();  
  
        controller.setStudentName("Prem");  
        controller.setStudentGrade("A+");  
  
        System.out.println("After Update");  
        controller.updateView();  
    }  
}
```

